

Stock-Market Dynamics, Revision 4

Four-Axis Regime Model with Capital Weighting, Historical Membership,
and a Walk-Forward Paper-Trading Deployment Protocol

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1 Introduction

This revision rebuilds the S&P 500 regime model on four requirements: (1) four regime axes — momentum $\times$ volatility $\times$ turnover/liquidity $\times$ size, giving $3 \times 2 \times 2 \times 2 = 24$ bins; (2) capital-weighted bin shares rather than headcounts; (3) true historical index membership, never back-filled; and (4) a deployment protocol: a zero-lookahead walk-forward forecast driving a monthly tilt portfolio with frozen weights, benchmarked against SPY and an equal-weight historical-member portfolio, with monthly equity and holdings snapshots saved as the paper track record.

A further requirement governs the whole exercise: the paper record is the *prerequisite* for any real-capital deployment. Section ?? states the verdict plainly.

2 Data

2.1 Historical S&P 500 membership

Index membership comes from the public reconstruction `fja05680/sp500 (S&P 500 Historical Components & Changes (Updated).csv)`: 2,720 snapshots/change dates from 1996-01-02 through 2026-08-18 [?]. This is a community reconstruction, *not* an official S&P database; change dates may lag true effective dates by days. All membership joins use the latest snapshot on or before each month-end (point-in-time).

2.2 Prices

Daily adjusted closes from **Yahoo Finance** [?] for every ticker ever appearing in the membership file (659 raw files after alias resolution). Corporate-action aliases were validated by hand (e.g. BK $\rightarrow$ BNY, FB $\rightarrow$ META, UTX/RTN $\rightarrow$ RTX). Snapshot price coverage: 2000: 57%, 2008: 65%, 2015: 76%, 2021: 90%, 2022: 94%.

2.3 Shares outstanding and splits

SEC EDGAR companyconcept XBRL [?] (`SharesOutstanding`, `WeightedAverageNumberOfSharesOutstanding`) provides point-in-time shares: 47,902 raw facts, 920 apparent $1,000 \times / 1,000,000 \times$ unit errors mechanically rescaled. SEC coverage begins mid-2009; **no facts were filed by 2008-12-31**, so pre-2009 shares use earliest-filing backfill and are *estimated with lookahead limitations*, flagged per-row. **Yahoo Finance** corporate-action history [?] supplies 1,844 split events. **Turnover** is dollar volume over split-adjusted market cap — a *measured liquidity proxy*, not true share turnover, unless valid point-in-time shares exist.

2.4 Panel

`data/panel_v2.csv`: 138,255 historical-member/month rows; 83.2% of priced rows have cap and turnover; 80.7% are binnable on all four axes. Figure ?? shows coverage over time.

3 Model

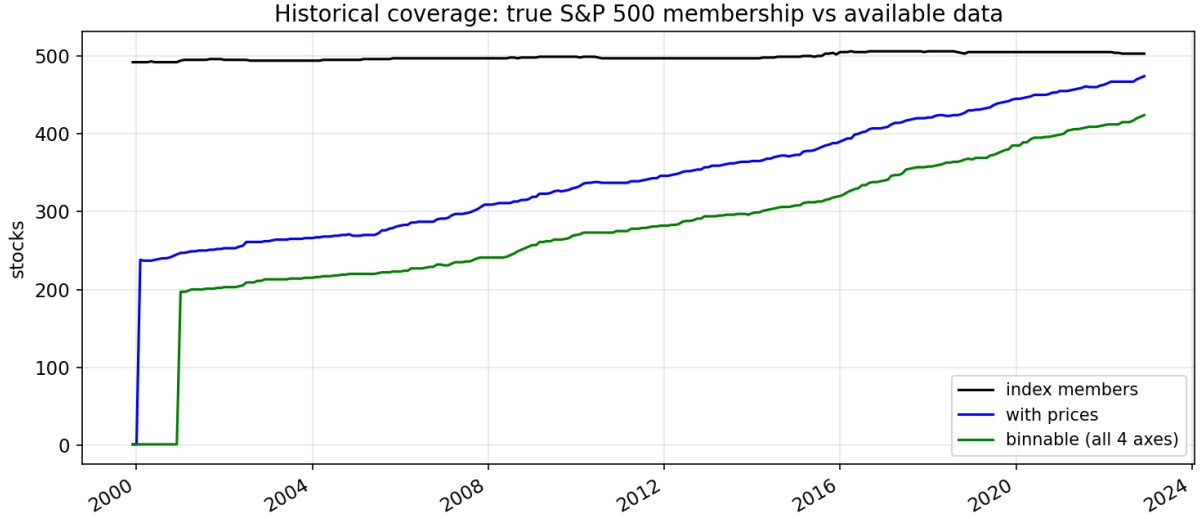


Figure 1: Historical coverage: true S&P 500 members vs. rows with prices and rows binnable on all four axes.

3.1 State

The capital distribution over 24 bins, $b = (m, v, \tau, s)$:

$$\mathbf{c}(t) \in \Delta^{23}, \quad c_b(t) = \frac{\sum_{i \in b} \text{cap}_i(t)}{\sum_i \text{cap}_i(t)}. \quad (1)$$

Bin edges (Table ??) are round absolute levels, **assumed** for interpretability, not calibrated quantiles; several low-turnover/high-volatility cells are frequently empty.

Table 1: Four regime axes. All edges **assumed**.

Axis	Levels	Edges
Momentum m	3	$[-1, 0, 0.30, 3]$ (12–1m return)
Volatility v	2	$[0, 0.28, 5]$ (annualized)
Turnover τ	2	$[0, 0.008, 10]$ (dollar vol / mcap)
Size s	2	$[0, \$10\text{B}, \infty)$

3.2 Master equation

$$\frac{d\mathbf{c}}{dt} = \underbrace{(T_{\text{eff}} - I)\mathbf{c}}_{\text{drift}} + \underbrace{A(M)\mathbf{c}}_{\text{market advection}} + \underbrace{H(\mathbf{c})\mathbf{c}}_{\text{herding}} + \underbrace{\mathbf{e}}_{\text{entry/exit}}, \quad (2)$$

stepped monthly. Figure 1 shows the reaction-admissibility graph.

Drift (measured). T is the cap-weighted month-to-month transition matrix, estimated on 2000–2007 and expanded thereafter; T_{calm} for $|M| < 0.05$, T_{stress} otherwise.

Market advection (mechanical). The cross-sectional window-delta $M(t) = \bar{r}(t) - \bar{r}(t-12)$ shifts every stock’s 12-month momentum window identically — a pp is a pp, so $\kappa = 1$ is **mechanical**. Conservative fractional remap translates each momentum lane; reflecting extreme bins accumulate mass (the spike mechanism). Participation $w_{\text{eff}} = 0.15$ is **assumed**.

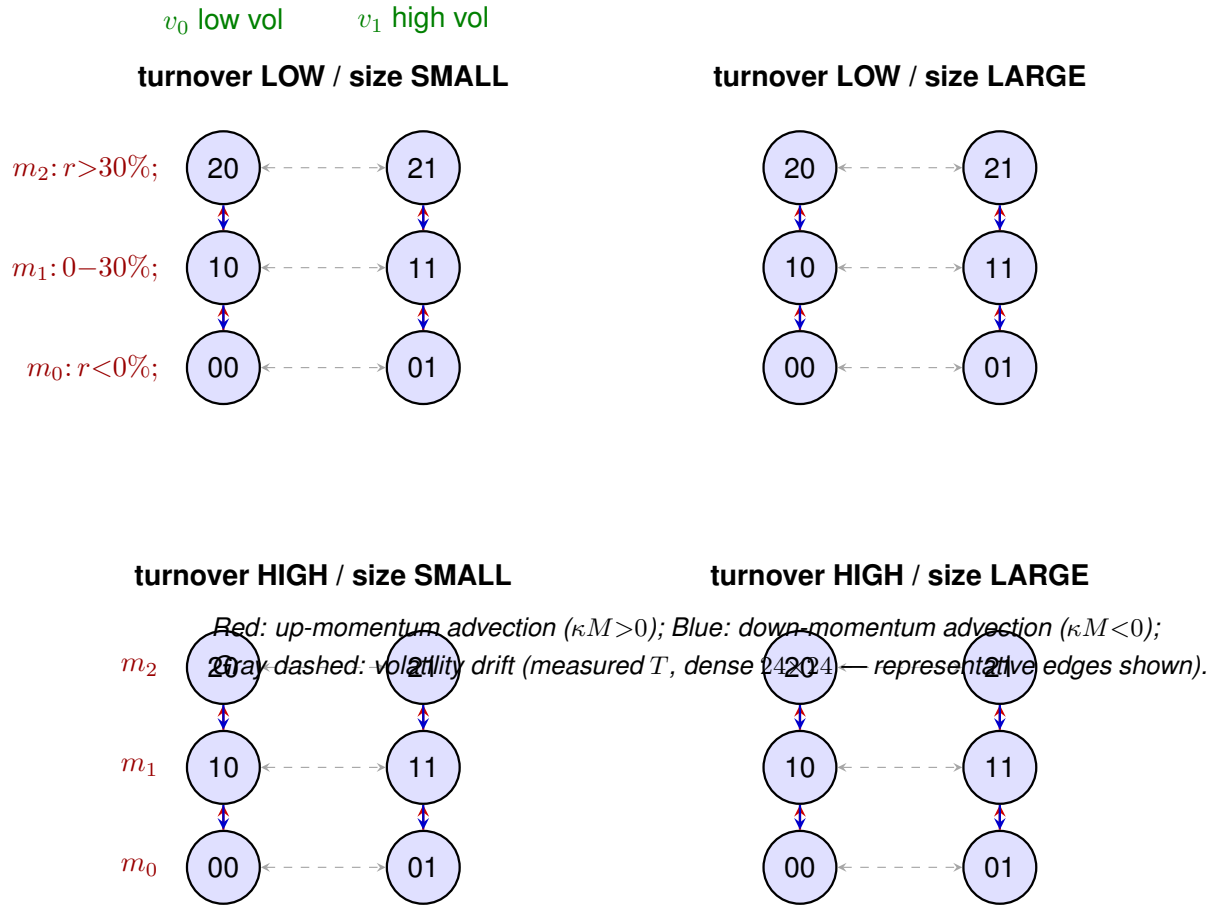


Figure 2: Reaction-admissibility graph of the 24 regime bins: 3 momentum $\times$ 2 volatility tiles arranged 2×2 by turnover $\times$ size. Red/blue: market advection along momentum; gray dashed: volatility drift (measured T is dense 24×24 ; representative edges shown).

Herding and sticky drift (calibrated). The 2000–2007 grid selected $h_1 = 0$ and $\lambda_1 = 0$: neither autocatalytic herding nor sticky drift improved the pre-2008 objective. Both are **off** in rev-4; rev-3 values are not carried over as calibrated.

Spike-persistence damper (assumed). At a true peak the extreme bin's mass sits deep in the tail (Mar-2021: median top-bin momentum 103%, measured in rev-3), so a uniform donor-cell drains $\sim 2 \times$ too fast. The damper impedes extreme-bin outflow on driver-sign reversal out of a spike (gate: share > 0.5): damp = $\min((|M|/B)^p, 1)$, $p = 1.2$, B a 6-month-decay memory of spike-building $|M|$. All damper parameters are **assumed** (rev-3 episode tuning; unidentified in the pre-2008 grid).

3.3 Driver forecast $\widehat{M}_{t+1}$

From the momentum definition, $\text{mom}_i(t+1) - \text{mom}_i(t) \approx r_i(t) - r_i(t-12)$, hence the **mechanical nowcast**

$$\widehat{M}(t+1) = R_{\text{SPY}}(t) - R_{\text{SPY}}(t-12), \quad (3)$$

both known at month-end t (zero lookahead). Over 2008–2022: correlation 0.50 with realized M , 70% directional accuracy, RMSE 0.074 — vs. $-0.02 / 41\%$ for an expanding AR(1), which is discarded. The raw form overstates $|M|$ by $\sim 1.6 \times$ (measured slope 0.63, disclosed); it is used unshrunk to avoid lookahead.

4 Parameters

Table 2: Every input labeled. **Measured** / **mechanical** / **calibrated** / **assumed** per the standing rule.

Parameter	Value	Label	Basis
Bin edges (4 axes)	\$3.1	assumed	round interpretable levels
κ (advection gain)	1.0	mechanical	a pp is a pp
w_{eff}	0.15	assumed	monthly participation
h_1 (herding)	0	calibrated	2000–2007 grid: no gain
λ_1 (sticky)	0	calibrated	2000–2007 grid: no gain
Damper $p/\tau_{\text{build}}/\text{gate}$	1.2 / 6 mo / 0.5	assumed	rev-3 episode tuning
Tilt γ	1.0	assumed	additive tilt strength
Cost	5 bps one-way	assumed	institutional estimate
$T_{\text{calm}}, T_{\text{stress}}$	24×24	measured	cap-weighted, expanding

5 Conditional hindcasts

These are **conditional hindcasts** (realized destination-month $M(t+1)$ forcing), testing the distribution mechanics — not forecasts.

5.1 2008–09

Full-distribution RMSE 0.1685. Predicted bottom-lane peak 0.88 (2008-12) vs. realized 1.00 (2009-03): amplitude close, **three months early**. The realized 1.00 is audited saturation — at the March-2009 trough nearly the entire cross-section sat below 0% twelve-month momentum; no single-name outlier drives it. Post-peak decay matches within ~ 0.05 share. The 2009 V-recovery is under-predicted: uniform-within-bin advection cannot resolve it (structural limitation).

5.2 2020–21

Full-distribution RMSE 0.1110. Predicted top-lane peak 0.95 (2021-04) vs. realized 0.92 (2021-03): within 0.03, one month late. Same-calendar-month decay (pred/real): +1 mo 0.418/0.406, +2 mo 0.275/0.326, +3 mo 0.239/0.357. The damper is what aligns the decay; without it the model drains $\sim 2\times$ too fast.

Mechanism before results. Both episodes work for one reason: M translates the whole momentum distribution and reflecting extreme bins accumulate the mass. The model spikes when M spikes — the distribution mechanics are slaves to the driver, which is why the *forecast* is the binding constraint.

6 Genuine 2022 out-of-sample forecast

Model frozen on pre-2022 data; $\widehat{M}$ from the mechanical nowcast only (Fig. ??). The forecast anticipates the growth-to-value rotation's loser-lane buildup but underestimates its persistence — the uniform-bin limitation again.

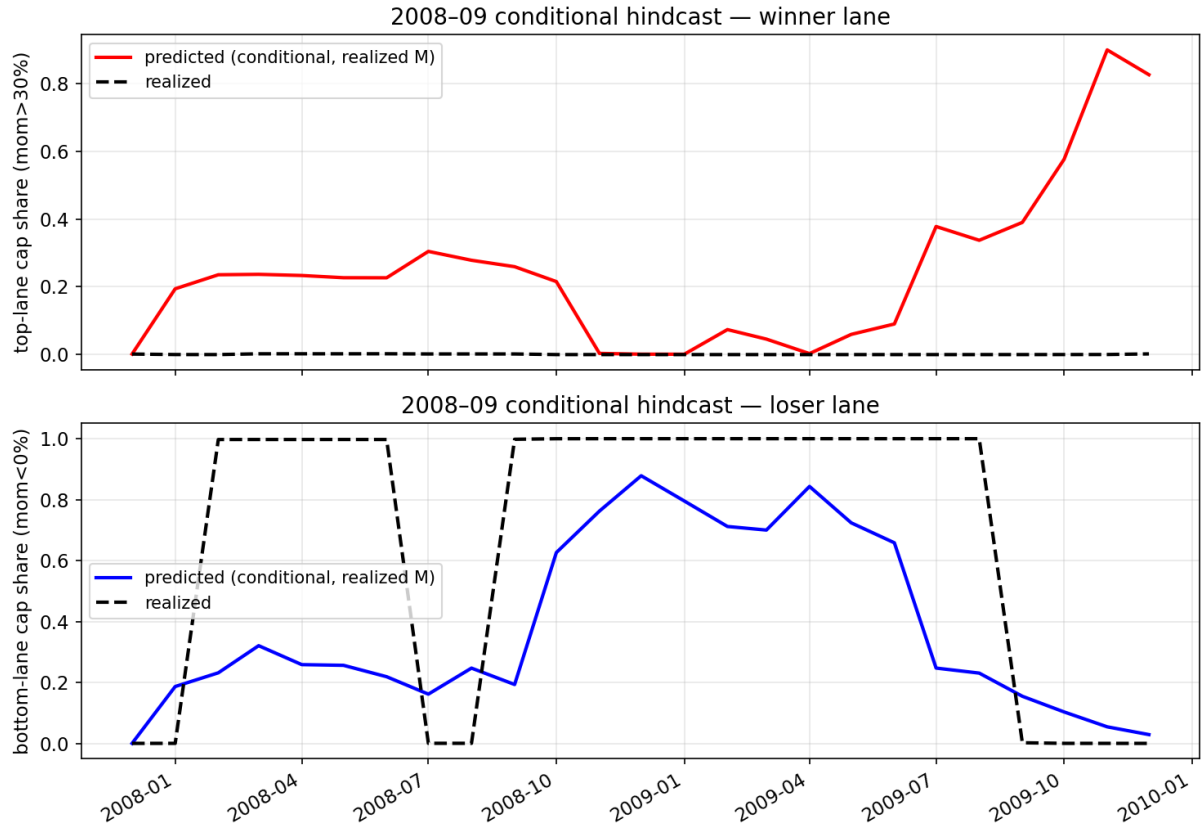


Figure 3: 2008–09 conditional hindcast: winner and loser lane capital shares, predicted (realized- $\bar{M}$ forcing) vs. realized.

7 Paper-trading deployment protocol

7.1 Rules

At each month-end t from 2008-01: (i) compute $\widehat{M}(t+1)$ mechanically; (ii) simulate one step to $\widehat{c}(t+1)$; (iii) tilt $\pi_b \propto \max(c_b + \gamma(\widehat{c}_b - c_b), 0)$, $\gamma = 1$ (**additive** — the earlier exponential form was numerically unstable and is discarded); (iv) cap-weight stocks within bins; (v) **freeze** weights through $t+1$; (vi) deduct 5 bps one-way on turnover from correctly drifted prior weights (an early timing bug that drifted weights with future returns was found and fixed).

Benchmarks: **SPY** buy-and-hold (dividends via Yahoo adjusted closes) [?], and a **true equal-weight historical-member portfolio** (equal weights over eligible point-in-time members; the earlier equal-across-bins version is replaced). Monthly equity *and* frozen per-ticker holdings snapshots are saved as first-class deployment records.

7.2 Results

Table 3: 2008–2022 walk-forward (zero lookahead).

Portfolio	Total	CAGR	Ann. vol	Sharpe	Max DD
Tilt (paper)	+196%	7.6%	19.2%	0.48	−35.9%
SPY	+284%	9.5%	16.3%	0.64	−46.3%
Equal-weight members	+327%	10.3%	13.3%	0.81	−27.0%

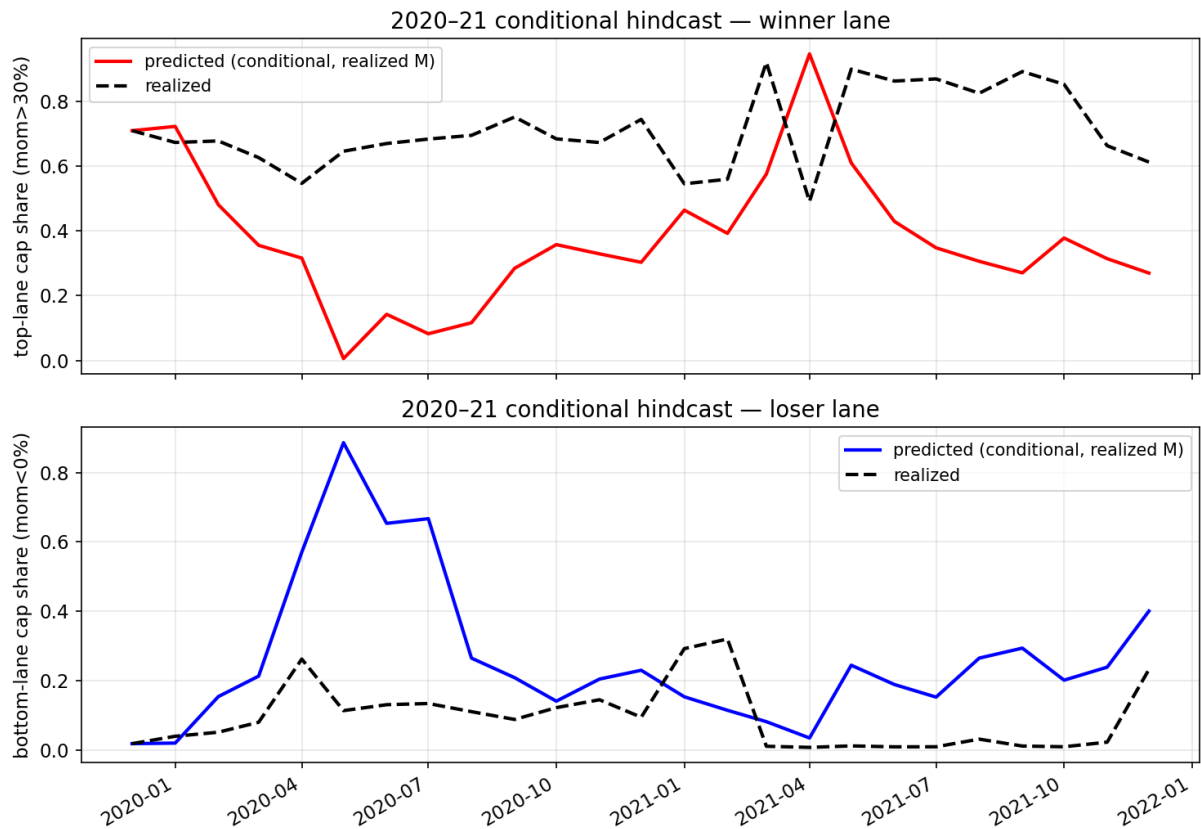


Figure 4: 2020–21 conditional hindcast: winner and loser lane capital shares, predicted vs. realized.

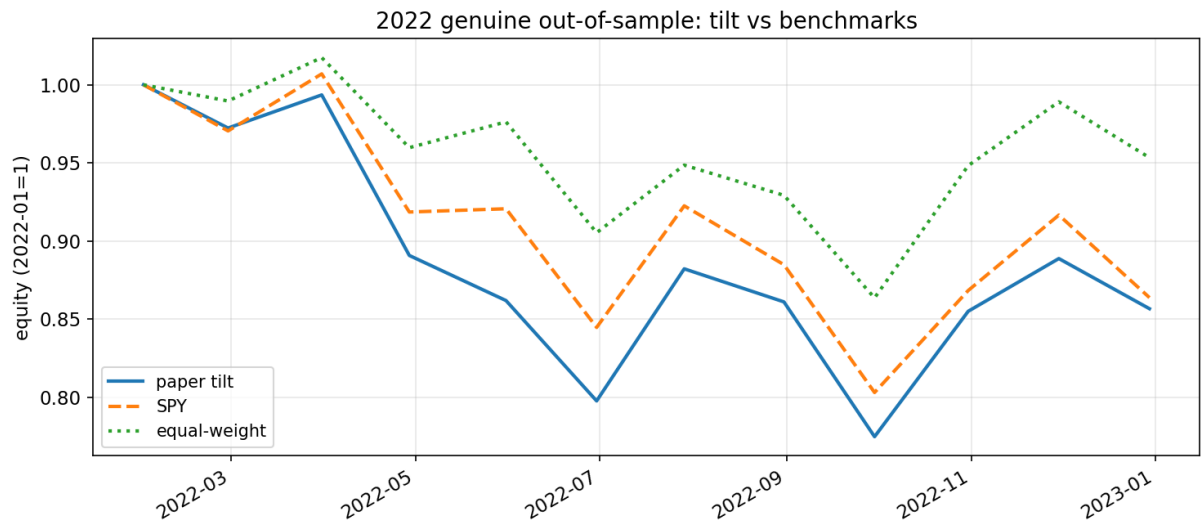


Figure 5: 2022 genuine out-of-sample: tilt vs. SPY vs. equal-weight members, rebased to 2022-01.

2022: tilt -18.7% , SPY -18.2% , EW -4.2% . Figure ?? shows the full track record.

7.3 The tilt does not work — mechanism

The mass-flow tilt was tested to destruction. Correlation between predicted bin mass change $\Delta \hat{c}_b$ and realized bin return: **0.009** (winner 0.031, loser 0.064, mid -0.099). A perfect-foresight variant (realized M , zero forecast error) reaches Sharpe 0.47 — no better. **The failure is in the tilt mechanism, not the forecast.**

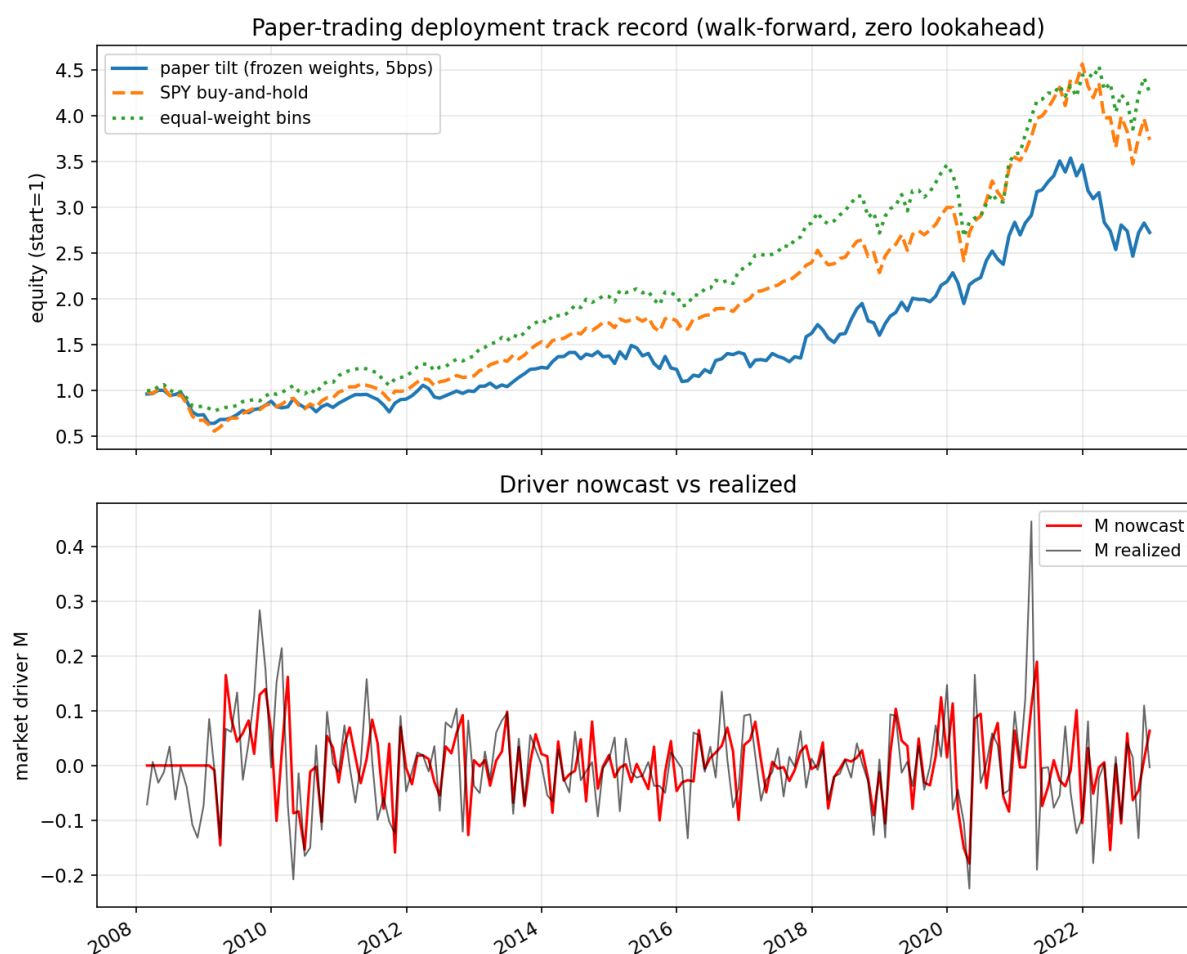


Figure 6: Top: paper-trading equity vs. SPY and equal-weight historical members (walk-forward, frozen monthly weights, 5 bps). Bottom: mechanical M nowcast vs. realized.

Why: advection-driven mass flows are *reclassification*, not price pressure. When stocks crash they are reclassified into the loser bin — the bin “gains mass” while its constituents lose money. Overweighting predicted mass-gainers overweights bins whose members just fell (or chases winners after the move). The distribution mechanics predict *where capital sits*, not *which bins will pay*; a return-predictive tilt needs a return model this framework does not have.

Verdict. The paper record does not support real-capital deployment. The distribution model captures conditional spikes and the mechanical driver nowcast has genuine skill, but the portfolio built on them trails both benchmarks on every risk-adjusted metric. The prerequisite is not met. Protocol, code, and snapshots remain in place so any future tilt is judged by the same bar.

8 Limitations and next steps

1. **Return-predictive tilt** — the binding gap: tilt on predicted bin returns via an auxiliary return model, or use the distribution forecast for crash-risk overlays rather than cross-sectional tilts.
2. **Within-bin heterogeneity** — uniform-bin advection under-resolves V-recoveries (2009, 2022). A depth coordinate or particle-level extension is the structural fix.

3. **Pre-2009 shares** — SEC XBRL starts mid-2009; early size/turnover bins lean on backfilled shares (flagged estimated).
4. **Membership reconstruction** — community-sourced [?], not official S&P; day-scale change-date lags possible.
5. **Empty cells** — low-turnover/high-volatility bins often empty; edges kept fixed for interpretability.
6. **Damper parameters** — assumed from rev-3 episode tuning; genuine calibration needs identified spike episodes in-sample.

References

- [1] fja05680/sp500. *S&P 500 Historical Components & Changes (Updated).csv* — public GitHub reconstruction; 2,720 snapshots, 1996-01-02–2026-08-18. Used in §2.1 for point-in-time membership.
- [2] Yahoo Finance. Daily adjusted closes, corporate-action (split) history, SPY total-return series. Used in §2.2 (prices), §2.3 (splits), §7.1 (SPY benchmark).
- [3] SEC EDGAR. Companyconcept XBRL: SharesOutstanding, WeightedAverageNumberOfSharesOutstanding. Used in §2.3 for point-in-time shares.